CAMBRIDGE
International Education

# Cambridge International AS & A Level

CANDIDATE NAME

CENTRE NUMBER

CANDIDATE NUMBER

*7222207883 9*

**MATHEMATICS** **9709/42**

Paper 4 Mechanics **May/June 2026**

**1 hour 15 minutes**

You must answer on the question paper.

You will need: List of formulae (MF19)

**INSTRUCTIONS**

- Answer **all** questions.
- Use a black or dark blue pen. You may use an HB pencil for any diagrams or graphs.
- Write your name, centre number and candidate number in the boxes at the top of the page.
- Write your answer to each question in the space provided.
- Do **not** use an erasable pen. Do **not** use correction fluid or tape.
- Do **not** write on any bar codes.
- If additional space is needed, you should use the lined page at the end of this booklet; the question number or numbers must be clearly shown.
- You should use a calculator where appropriate.
- You must show all necessary working clearly; no marks will be given for unsupported answers from a calculator.
- Give non-exact numerical answers correct to 3 significant figures, or 1 decimal place for angles in degrees, unless a different level of accuracy is specified in the question.
- Where a numerical value for the acceleration due to gravity ($g$) is needed, use $10\,\text{m}\,\text{s}^{-2}$.

**INFORMATION**

- The total mark for this paper is 50.
- The number of marks for each question or part question is shown in brackets [ ].

This document has **16** pages. Any blank pages are indicated.


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**1** A car of mass 960 kg is moving on a straight horizontal road. There is a constant force of magnitude 620 N resisting the motion of the car.

**(a)** Calculate the power developed by the engine of the car when it is moving at a constant speed of $25\,m\,s^{-1}$. [2]

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**(b)** Given that the power is suddenly increased **by** 12 kW, find the instantaneous acceleration of the car. [2]

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**2** Two particles, $A$ and $B$, of masses 2.5 kg and 3.5 kg respectively are at rest on a straight smooth horizontal track. Particle $B$ is situated 9 m from a vertical wall which is fixed at right angles to the track.

Particle $A$ is projected directly towards $B$ with a speed of $3\,\text{m}\,\text{s}^{-1}$. In the subsequent motion, $A$ collides and coalesces with $B$ to form particle $C$. Particle $C$ then collides directly with the wall and rebounds. The collision of $C$ with the wall reduces the speed of $C$ by 25%.

**(a)** Find the speed of $C$ after it rebounds from the wall. [2]

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**(b)** Hence find the time from the instant at which $A$ and $B$ collide until $C$ is once again a distance of 9 m from the wall. [2]

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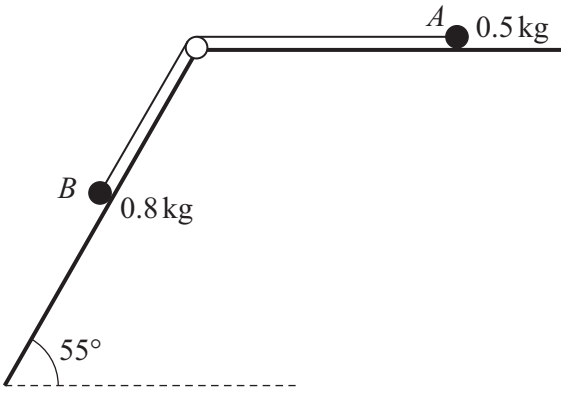
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Two particles, $A$ and $B$, of masses 0.5 kg and 0.8 kg respectively are attached to the ends of a light inextensible string. The string passes over a small smooth pulley fixed at the end of a rough horizontal plane and to the top of a smooth inclined plane. Particle $A$ is held on the horizontal plane, while $B$ lies on the inclined plane, which makes an angle of 55° with the horizontal. The string is in the same vertical plane as a line of greatest slope of the inclined plane (see diagram).

Particle $A$ is released from rest with the string taut. Particle $B$ moves 0.3 m down the inclined plane in 0.4 s.

**(a)** Find the tension in the string. [4]

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**(b)** Find the coefficient of friction between $A$ and the horizontal plane. [3]

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**(c)** After $B$ has been moving for 0.4 s, the string suddenly breaks. Given that $A$ subsequently comes to rest on the horizontal plane, find the work done by the frictional force in bringing $A$ to rest. You may assume that $A$ does not reach the pulley. [3]

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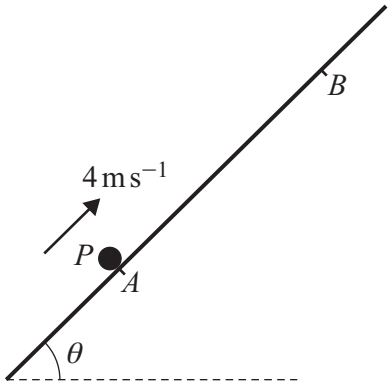
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**4**




A particle $P$ is projected from a point $A$ with speed $4\,\text{m}\,\text{s}^{-1}$ up a line of greatest slope of a rough plane inclined at an angle $\theta$ to the horizontal, where $\tan\theta = \frac{4}{3}$.

$P$ comes to instantaneous rest at a point $B$ on the plane (see diagram).

The coefficient of friction between $P$ and the plane is $\frac{1}{3}$.

**(a)** Using an energy method, show that the distance $AB$ is $0.8\,\text{m}$. [4]

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**(b)** After coming to instantaneous rest at $B$, the particle slides down the plane.

Find the total time from the instant at which $P$ is projected from $A$ until it returns to $A$. [5]

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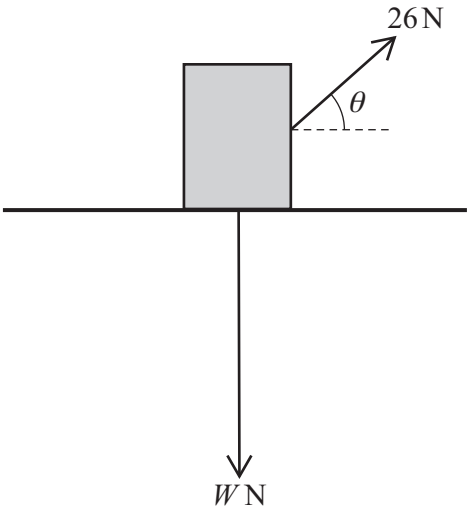
**5** The points $A$ and $B$ are at the same vertical height $h$ m above horizontal ground. A particle $P$ is released from rest from $A$. One second later, a particle $Q$ is projected vertically downwards from $B$ with speed $18\,\text{m}\,\text{s}^{-1}$.

Given that $P$ and $Q$ reach the ground at the same time, find the value of $h$. [6]

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**6**




A crate of weight $W$ N is at rest on rough horizontal ground. The crate is pulled at a constant acceleration of $1.5\,\text{m}\,\text{s}^{-2}$ along the ground in a straight line by a light rope. The rope is inclined at an angle of $\theta$ to the horizontal, where $\theta = \sin^{-1}\frac{5}{13}$. The tension in the rope is 26 N (see diagram). The coefficient of friction between the crate and the ground is 0.25.

**(a)** Find the value of $W$. [5]

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**(b)** When the crate reaches a point $A$, the rope is removed. The speed of the crate at $A$ is $8\,\text{m}\,\text{s}^{-1}$.

The crate comes to rest at a point $B$.

Find the distance $AB$. [3]

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**7** A particle $P$ moves in a straight line. The velocity $v\,\text{m}\,\text{s}^{-1}$ of $P$ at time $t\,\text{s}$, where $t \geqslant 0$, is given by

$$v = k_1(4t+11)^{\frac{1}{2}} - \tfrac{1}{4}(2t+1)^2 + k_2,$$

where $k_1$ and $k_2$ are constants. When $t = 1.25$ the deceleration of $P$ is $0.5\,\text{m}\,\text{s}^{-2}$.

**(a)** Find the value of $k_1$. [4]

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**(b)** Given that $P$ is only at instantaneous rest when $t = 3.5$, find the distance travelled by $P$ in the interval $0 \leqslant t \leqslant 1$. [5]

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**Additional page**

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